

Pratik Vijay Jadhav

📍 Pune, Maharashtra  LinkedIn  Github  pratik.ja3456@gmail.com  +91 9021130978

SKILLS

- **Programming Languages:** Python, R, SQL
- **Frameworks & Tools:** FastAPI, Flask, Streamlit, Selenium, BeautifulSoup, OpenCV, TensorFlow, Keras, Scikit-Learn, NumPy, Pandas, Docker, Celery, RabbitMQ, Plotly
- **Machine Learning & AI:** Neural Networks, Transfer Learning, Computer Vision, NLP, LLMs (RAG, RAFT fine-tuning), Hyperparameter Tuning, MLOps
- **Databases:** MySQL, MongoDB
- **Internet of Things (IoT):** Arduino, ESP-32, Edge & Cloud Computing

EDUCATION

Atria University, Bachelor of Technology (B. Tech) in Computer Science (87.6%) 11/2021 – 08/2025 | Bangalore, Karnataka

- **Specialization:** Digital Transformation
- **Relevant Coursework:** Machine Learning, Deep Learning, Data Science, Data Wrangling & Visualization, Database Management, Big Data and MLOps, Computer Vision, NLP Specialization.

PROFESSIONAL EXPERIENCE

Alemeno Pvt. Ltd., Machine Learning Intern

- Built an **AI defect detection assistant** for LCD panels by integrating **LLMs (Claude, GPT, Gemini) and CNNs**, improving classification accuracy from ~34% to **92%** across **15+ defect types**.
- Engineered **large-scale datasets (10000+ images)** with preprocessing, context images, and multi-defect detection logic; applied iterative **prompt optimization & fine-tuning (Qwen 2.5-VL)** to boost accuracy on challenging defects to **70–90%**.
- Deployed the system as a **production-ready pipeline** with **GCP AutoML + TensorFlow Serving (Dockerized)** and an **ensemble LLM+CNN inference model** for real-time batch predictions.

Joshnik AI Labs, AI & Data Engineering Intern

- Collected and processed **50k+ professional profiles** (doctors, plumbers, trainers, etc.) from **40+ categories** using **Selenium, Crawl4AI, Google Places API**, and enriched them with reviews, ratings, and geo-coordinates for structured search.
- Built an **AI-powered discovery chatbot** that interprets user queries in natural language, asks follow-up questions, and retrieves relevant professionals using **LLMs (OpenAI, Claude, Gemini) + FAISS vector search (sentence-transformers)**.
- Supported deployment with a **Dockerized FastAPI + MongoDB backend, Celery + RabbitMQ async processing, and Next.js dashboard**, enabling real-time search, Google Maps directions, and WhatsApp/DM integration for booking.

Cheslab Private Limited, Machine Learning and App Development Intern 

- Mapped and prototyped wireframes for a **skincare diagnosis application**, creating a streamlined onboarding process that reduced user **drop-off by 15%** and improved overall user experience.
- Curated a high-quality dataset of **100,000 labeled and 50,000 unlabeled skin condition images** with data preprocessing.
- Programmed and optimized a **CNN model in Python** for skin disease classification.

Research Assistant – Agriculture LLM for Smart Farming, Atria University | Research Assistant under Prof. Dinker Mattam

- Conducted research on **LLM-based agricultural assistants**, implementing **RAG** for real-time agricultural insights and **RAFT fine-tuning**, improving response accuracy by **30%**.
- Implemented a **robust RAG pipeline** to enhance document retrieval, leading to more contextually relevant AI-generated responses.
- Created a **judge model to evaluate RAG-based responses** by **fine-tuning a LLaMA model**, ensuring higher response quality and reliability.
- Built an **interactive UI with Streamlit**, allowing users to seamlessly interact with the model and receive AI-driven farming insights.

PROJECTS

Music Recommendations based on Facial Emotion Recognition

- Engineered a music recommendation system using facial emotion recognition with **Streamlit UI**, achieving **85% accuracy** in mood detection with the **FER-2013 dataset**, and integrated with **Spotify API** for personalized playlists.

Audio Caption Generation System for the Visually Impaired

- Developed a **real-time platform** for visually impaired users, generating spoken image descriptions using **VGG-16 and Transformer model**, achieving **87% accuracy** on benchmark captioning datasets..
- Integrated **Pyttsx3** for text-to-speech conversion and deployed a **Flask-based API** for real-time accessibility support.

CERTIFICATES

- Natural Language Processing, Coursera (Completed: 02/2025) 
- MLOps | Machine Learning Operations, Coursera (Completed: 10/2024) 
- Cloud Security, Palo Alto Networks Cybersecurity Academy (Completed: 06/2024) 